

KEGG KEMET module and metabolic pathway	Module status
MD0001 Gluconogenesis, oxalacetate→fructose 6P	Complete
MD0004 Pentose phosphate pathway/Pentose phosphate cycle	Complete
MD0005 PPPP biosynthesis, ribose 5P→PPP	Complete
MD0006 Pentose phosphate pathway, oxidative phase, glucose 6P→fructose 5P	Complete
MD0007 Pentose phosphate pathway, non-oxidative phase, fructose 6P→fructose 1P	Complete
MD0008 Entner-Doudoroff pathway, glucose 6P→glyoxalaldehyde 3P + pyruvate	Complete
MD0009 Citrate cycle (TCA cycle, Krebs cycle)	Complete
MD0010 Citrate cycle, first carbon oxidation, oxalosuccinate→2-oxoglutarate	Complete
MD0011 Citrate cycle, second carbon oxidation, 2-oxoglutarate→oxalosuccinate	Complete
MD0012 Glyoxylate cycle	Complete
MD0013 Malonate semialdehyde pathway, propionyl-CoA→acetyl-CoA	Complete
MD0014 Gluconate pathway/Luxonate pathway	Complete
MD0015 Pyruvate biosynthesis, glutamate→pyruvate	Complete
MD0016 Lysine biosynthesis, succinyl-CoA pathway, aspartate→lysine	Complete
MD0017 Methionine biosynthesis, aspartate→homoserine→methionine	Complete
MD0018 Threonine biosynthesis, aspartate→homoserine→threonine	Complete
MD0019 Valine/isoleucine biosynthesis, pyruvate→valine / 2-oxoglutarate→isoleucine	Complete
MD0020 Serine biosynthesis, glycine→3P→serine	Complete
MD0021 Cysteine biosynthesis, serine→cysteine	Complete
MD0022 Isoleucine pathway, glyceraldehyde 3P→pyruvate + acrylylase 4P→chrysinate	Complete
MD0023 Tyrosophenyl biosynthesis, chorismate→tyrosophenyl	Complete
MD0024 Phenylalanine biosynthesis, chorismate→phenylalanine	Complete
MD0025 Tyrosine biosynthesis, chorismate→tyrosine	Complete
MD0026 Isoleucine biosynthesis, PMP→tyrosine	Complete
MD0027 CABA (lysine-β-oxalobutyrate) shunt	Complete
MD0028 Ornithine biosynthesis, glutamate→ornithine	Complete
MD0029 Urea cycle	Complete
MD0030 Lysine biosynthesis, AAA pathway, 2-oxoglutarate→2-aminoadipate→lysine	Complete
MD0031 Lysine biosynthesis, mediated by LysW, 2-aminoadipate→lysine	Complete
MD0032 Lysine degradation, lysine→acetyl-CoA→acetyl-CoA→acetyl-CoA	Complete
MD0033 Ethanol biosynthesis, aspartate→ethanol	Complete
MD0034 Methionine salvage pathway	Complete
MD0035 Methionine degradation	Complete
MD0036 Leucine degradation, leucine→acetoacetate + acetyl-CoA	Complete
MD0037 Melatonin biosynthesis, tryptophan→serotonin→melatonin	Complete
MD0038 Tryptophan metabolism, tryptophan→tryptamine→5-aminotryptamine	Complete
MD0039 Melanogel biosynthesis, phenylalanine/tyrosine→melanogel	Complete
MD0040 Tyrosine biosynthesis, tyrosine→tyrosine→tyrosine	Complete
MD0042 Catecholamine biosynthesis, tyrosine→dopamine→noradrenaline→adrenaline	Complete
MD0043 Thyroid hormone biosynthesis, tyrosine→3-iodothyronine/tyrosine	Complete
MD0044 Tyrosine degradation, tyrosine→glutamate	Complete
MD0045 Tyrosine degradation, histidine→3-oxoindolylpyruvate→glutamate	Complete
MD0046 Pyrimidine degradation, uracil→beta-alanine, thymine→3-aminobutanoate	Complete
MD0047 Creatine pathway	Complete
MD0048 Isoleucine monophosphate biosynthesis, PRPP + glutamine→IMP	Complete
MD0049 Adenine ribosuccinide biosynthesis, IMP→ADP/ATP	Complete
MD0050 Guanine ribosuccinide biosynthesis, IMP→ADP/ATP	Complete
MD0051 Uridine monophosphate biosynthesis, glutamine + PRPP→UMP	Complete
MD0052 Pyrimidine ribosuccinide biosynthesis, UMP→UDP/UTP, CDP/CTP	Complete
MD0053 Pyrimidine deoxyribosuccinide biosynthesis, CDP/CTP→dCDP/dCTP, dTDP/dTTP	Complete
MD0055 N-glycine precursor biosynthesis	Complete
MD0056 D-glycine biosynthesis, much type core	Complete
MD0057 Glutamine biosynthesis, lysine→methionine	Complete
MD0058 Glutamine biosynthesis, glutamate→glutamine	Complete
MD0059 Glutamine biosynthesis, glutamate→glutamine	Complete
MD0060 KDQ2-lipid A biosynthesis, Raetz pathway, LpA, LpA type	Complete
MD0061 D-glucuronate degradation, D-glucuronate→pyruvate + D-glyoxalaldehyde 3P	Complete
MD0062 CWP-KDS	Complete
MD0064 D-glucuronate degradation, D-glucuronate→pyruvate	Complete
MD0065 Glutamine biosynthesis, core oligosaccharide	Complete
MD0066 Lactochylceramide biosynthesis	Complete
MD0067 Sulfolipid biosynthesis, ceramide3-alkyl-2-acylglycerol→sulfolipid/serinolipid	Complete
MD0068 Glycosphingolipid biosynthesis, glybo-series, LacCer→Gb4Cer	Complete
MD0069 Glycosphingolipid biosynthesis, ganglio-series, LacCer→Gb4Cer	Complete
MD0070 Glycosphingolipid biosynthesis, lacto-series, LacCer→Gb4Cer	Complete
MD0071 Glycosphingolipid biosynthesis, melacto-series, LacCer→Gb4Cer	Complete
MD0072 N-glycosylation by oligosaccharyltransferase	Complete
MD0073 N-glycan precursor trimming	Complete
MD0074 N-glycan biosynthesis, high mannose type	Complete
MD0075 N-glycan biosynthesis, complex type	Complete
MD0076 Dermatan sulfate degradation	Complete
MD0077 Chondroitin sulfate degradation	Complete
MD0078 Heparan sulfate degradation	Complete
MD0079 Keratan sulfate degradation	Complete
MD0080 Dermatan degradation	Complete
MD0081 Fatty acid biosynthesis, initiation	Complete
MD0083 Fatty acid biosynthesis, elongation	Complete
MD0085 Fatty acid elongation in mitochondria	Complete
MD0086 beta-Oxidation, acyl-CoA synthase	Complete
MD0087 beta-Oxidation	Complete
MD0088 beta-Oxidation, acyl-CoA synthase	Complete
MD0089 Triacylglycerol biosynthesis	Complete
MD0090 Phosphatidylcholine (PC) biosynthesis, choline→PC	Complete
MD0091 Phosphatidylcholine (PC) biosynthesis, PE→PC	Complete
MD0092 Phosphatidylethanolamine (PE) biosynthesis, ethanolamine→PE	Complete
MD0093 Phosphatidylethanolamine (PE) biosynthesis, PA→PE→PE	Complete
MD0094 Ceramide biosynthesis	Complete
MD0095 C3 isoprenoid biosynthesis, mevalonate pathway	Complete
MD0096 C3 isoprenoid biosynthesis, non-mevalonate pathway	Complete
MD0097 beta-Carotene biosynthesis, GGAP→beta-carotene	Complete
MD0098 Aracylgene degradation	Complete
MD0099 Lipopolysaccharide biosynthesis	Complete
MD0100 Lipopolysaccharide degradation	Complete
MD0101 Cholesterol biosynthesis, squalene 2,3-epoxide→cholesterol	Complete
MD0102 Ergosterol biosynthesis	Complete
MD0103 Cholesteriferol biosynthesis	Complete
MD0104 Beta acid biosynthesis, cholesterol→cholesteriferol/cholesterol	Complete
MD0106 Steroid-like acid biosynthesis, cholesterol→cholesteriferol/cholesterol	Complete
MD0107 Steroid hormone biosynthesis, cholesterol→pregnenolone→progesterone	Complete
MD0108 C21 Steroid hormone biosynthesis, progesterone→corticosterone/prednisolone	Complete
MD0109 C21 Steroid hormone biosynthesis, progesterone→corticosterone	Complete
MD0110 C19/C18 Steroid hormone biosynthesis, progesterone→androstenedione→testosterone	Complete
MD0112 Steroid alcohol biosynthesis	Complete
MD0113 Jovanone and biosynthesis	Complete
MD0114 Ascorbate biosynthesis, plenti, glucose 6P→ascorbate	Complete
MD0115 NAD biosynthesis, aspartate→NAD	Complete
MD0116 Menadiquinone biosynthesis, chorismate→menaquinol	Complete
MD0117 Ubiquinone biosynthesis, pre-ubiquinone, chorismate→ubiquinone	Complete